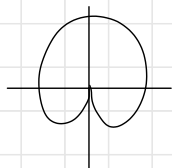


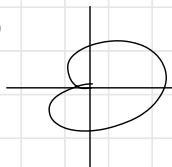
Cardioides

1)

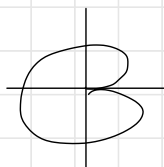


$$R = 1 + 5 \cos \theta \quad \text{en general } R = a + a \cos \theta$$

2)

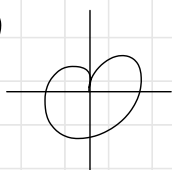


$$R = a + a \cos \theta$$



$$R = a - a \cos \theta$$

3)



$$R = a - a \sin \theta$$

Area en coordenadas Polares

$$A = \frac{1}{2} \int_a^b R^2$$

* $a \cos / \sin(\theta)$
 a solo cambia el radio

$\cos / \sin(a\theta)$
si a es par / el doble de pétalos
si a impar / la misma cantidad de pétalos

Longitud de arco

$$L = \int_a^b \sqrt{R^2 \left(\frac{dR}{d\theta} \right)^2} \cdot d\theta$$

Ecuaciones Paramétricas

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} \cdot dt$$